Cross-Attention-Driven Adaptive Graph Relational Network for Multilabel Remote Sensing Scene Classification

Haixia Bi, *Member, IEEE*, Honghao Chang, Xiaotian Wang, and Danfeng Hong, *Senior Member, IEEE*

***Abstract*— Multilabel remote sensing scene classification (MLRSSC) has garnered growing attention in recent years, owing to its more comprehensive description of land covers compared to its single-label counterpart. However, challenges arise inevitably. First, the relations among multiple scene labels are sophisticated. How to excavate the interclass dependencies is, therefore, a key challenge for the MLRSSC task. Second, extracting discriminative semantic features is essential, yet challenging for scene prediction of remote sensing images. Another issue is that the multilabel dataset usually shows twofold sample imbalances, that is, class imbalance and positive–negative imbalance, which have not been explored in MLRSSC tasks so far. To overcome the above hurdles, we put forward a cross-attention-driven adaptive graph relational network for the MLRSSC task. Different from the chain-like long short-term memory (LSTM) or static label co-occurrence matrices, we propose to use image-specific relational graphs to dynamically model the interclass dependencies. We innovatively devise a cross-attention-driven representation learning approach, which uses learnable label embeddings to query the class-wise semantic features, explicitly establishing the feature-label connections. Moreover, we design a balanced focal loss (BFL) function, where the loss contributions of positive and negative samples are rebalanced based on the respective imbalance degrees of diverse classes. Extensive experiments were performed on UCM, AID, and DFC15 multilabel datasets. Experimental results demonstrated that our proposed method achieves state-of-the-art performance in the studied task.**

***Index Terms*— Cross-attention, graph convolutional networks (GCNs), label dependency, multilabel classification, remote sensing.**

I. Introduction

BENEFITING from the booming development of remote sensing techniques, recent years have witnessed an unprecedented increase of remote sensing data volumes, which opens up new opportunities for researchers in the remote sensing community [1]. Remote sensing image scene classification, which aims to assign scene-level class labels to each image, plays an indispensable role among all remote sensing image processing tasks [2], [3]. It has been widely applied and holds promising prospects in diverse applications, such as forest management, sea monitoring, and urban planning [4], [5], [6], [7], [8].

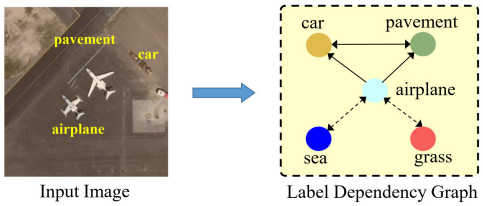


Fig. 1. Illustration of label dependency modeling. In the label dependency graph, the solid lines indicate strong class correlations, while the dashed lines signify weak class relations.

Early remote sensing image scene classification mainly focuses on single-label scenarios [9], where each image is assumed to belong to one single class. Exhaustive researches have been conducted on the above task [10], [11], [12], [13]. However, remote sensing images, especially aerial and satellite images, usually portray more than one scene, owing to their intrinsically expansive coverage of imaging. Fig. 1 gives three examples of aerial images with the presence of multiple objects. Taking the right-most image, for instance, it contains a variety of scenes, including *bare soil*, *buildings*, *grass*, and *pavement*, all of which are essential for understanding land cover. Under this circumstance, multilabel remote sensing scene classification (MLRSSC) task has emerged [14], whose objective is to assign multiple scene labels to an input image [15]. This task enables comprehensive and fine-grained semantic delineation of the Earth's surface, which is exactly the focus of this article.

Owing to the advantages and promises MLRSSC task holds, it has received widespread concerns in recent years [16]. Specifically, the explosive development of deep learning (DL) has significantly promoted the performance of the studied task [17], [18], [19], [20], [21], [22], [23], [24], [25]. Despite the remarkable progress, MLRSSC task is still confronted with several challenges, impeding their practical applications.

Label dependency modeling is the key challenge for the MLRSSC task. Some existing methods formulate the

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Haixia Bi and Honghao Chang are with the School of Information and Communications Engineering, Xi'an Jiaotong University, Xi'an 710049, China (e-mail: haixia.bi@xjtu.edu.cn; 2193712652@stu.xjtu.edu.cn).

Xiaotian Wang is with the Unmanned System Research Institute, Northwestern Polytechnical University, Xi'an 710072, China (e-mail: xtwang0111@nwpu.edu.cn).

Danfeng Hong is with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100094, China, and also with the School of Electronic, Electrical and Communication Engineering, University of Chinese Academy of Sciences, Beijing 100049, China (e-mail: hongdf@aircas.ac.cn).

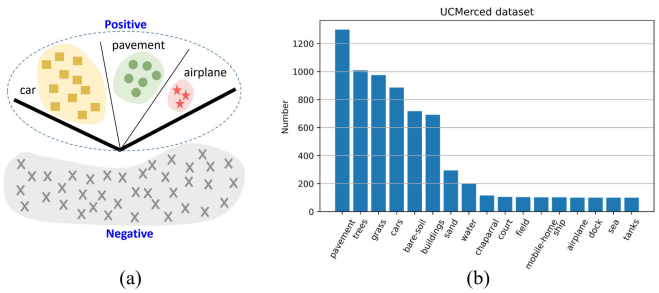
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multilabel classification problem as a set of binary classification problems and predict the existence of each scene class independently [26], [27]. However, the classification performance and generalization ability of these approaches have proved inferior due to the ignorance of label correlations [28]. Actually, for the multilabel classification problem, the output set of labels is inherently correlated. As illustrated in Fig. 1, the presence of *airplane* indicates a high co-occurrence probability of *car* and *pavement*, whereas is less likely to infer the existence of *sea* and *grass*. Incorporating the label dependency information into the classification would benefit the correct derivation of multiple labels, which is in perfect line with real-world perceptions [29]. Our motivation just originates from this simple, yet essential design philosophy. Investigating existing MLRSSC methods, we can discover that some works predict multiple scene labels in a sequential way using the recurrent neural network (RNN) or long short-term memory (LSTM) [14], [30], [31]. Nevertheless, this chain propagation manner focuses on correlations among neighboring labels, yet has limited ability in modeling graph-structured relationships. Furthermore, it is prone to error accumulation during the propagation process. The graph neural network (GNN), which is naturally advantageous in establishing interclass dependencies, is therefore introduced to the MLRSSC task. However, most graph-based methods derive label correlation graphs via statistically counting the co-occurrence frequencies or computing word embedding similarities between labels [14], [32], [33]. The label correlation graphs constructed in this manner are static and shared by all images, which may result in biased predictions for scenes that co-occur less frequently. Moreover, the label correlations of training and test sets are not completely consistent, where the static graphs may corrupt the generalization of the model [34]. Therefore, in this work, we propose to exploit dynamic graph for label dependency modeling, which adaptively constructs image-specific interclass relational graphs from the class-wise features. This graph establishment method fully considers the specificity of each image, where the built graphs represent unique content-aware category relations.

Semantic class representation learning is pivotal for graph construction and accurate object identification, which is, however, challenging for remote sensing images, as they usually contain complex spatial contextual information. Some researches [19], [35] first extract features with deep convolutional encoders such as VGG16 [36], [37] or ResNet [38] and then classify the scenes with multilabeling layers. These methods do not involve explicit guidance from labels in the training process, where the underlying sophisticated relations between features and labels cannot be fully mined and exploited. Another line of works [39], [40] resort to class attention modules to locate semantic regions, using which the feature-label relations are obtained. However, the class attention maps are generated via performing global average pooling on the last feature map, which discards the rich spatial information contained in the convolutional features [41]. Therefore, inspired by DETR [42], we develop a cross-attention-driven semantic class representation learning approach in this work. To be specific, we use learnable label embeddings as queries to examine the presence of land-cover objects by performing multihead cross-attention in an end-to-end manner. This enables accurate locating of class-wise semantic regions, thereby achieving effective extraction of semantic class representations.



(a) (b)

Fig. 2. Illustration of two sample imbalance issues in the MLRSSC task. (a) Positive–negative imbalance. (b) Class imbalance.

Apart from the above two challenges, *sample imbalance* is another longstanding, yet rarely investigated problem in the MLRSSC task. Most existing works directly use binary cross-entropy (BCE) loss function [14], [32], [43], [44] as the optimization objective. However, this loss function neglects two types of sample imbalances, that is, positive–negative imbalance and class imbalance, respectively. First, in the typical multilabel scenario, one image usually encompasses few positive labels and a large number of negative ones, indicating that the positive–negative ratio per class is on average low, as illustrated in Fig. 2(a). This would lead to overemphasis of the negative labels during optimization and further result in poor accuracy. Second, statistically analyzing all classes in the dataset, the positive–negative sample ratios vary from each other as well. As shown in Fig. 2(b), a typical long-tailed class distribution can be observed on the UCM dataset [45]. In this scenario, the trained model tends to be biased toward head classes with massive samples, resulting in inferior performance on tail classes with fewer samples. Although focal loss (FL) [46] was proposed to tackle the positive–negative sample imbalance issue, it does not take the class imbalance problem into account. To address this issue, we design a balanced FL (BFL) for multilabel classification. This loss function is cost-sensitive, which assigns different weights on positive and negative samples according to the imbalance degrees with respect to different classes.

The main contributions of this article are as follows.

1) We put forward a cross-attention-driven adaptive graph relational network, which we call CAGRN, for MLRSSC. To be specific, we propose to exploit dynamic graphs for label dependency modeling, which adaptively constructs image-specific interclass relational graphs. Cross-attention mechanism is exploited to extract class-wise semantic representations for graph construction, in which label embeddings are used as direct queries to locate scene positions, promoting the semantic representation ability and interpretability of features.
2) To simultaneously address the twofold sample imbalance issues, we design a new multilabel classification loss

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function, that is, BFL. It amplifies the influence of tail classes and positive labels with a class-relevant controlling mechanism, which is a generic and more flexible form of FL.

3) We conducted extensive experiments on three public remote sensing multilabel datasets, that is, UCM, AID, and DFC15 datasets. Results show that our proposed CAGRN achieves state-of-the-art performance compared with existing competitors.

The remainder of this article is organized as follows. Section II briefly recalls the related works. The proposed methodology is delineated in Section III. Section IV reports the comparative study on three benchmark multilabel datasets. Section V concludes the article and looks into future research prospects.

II. Related Works

We will review relevant works in the MLRSSC field from the following aspects: multilabel classification in remote sensing, label dependency modeling and attention-based methods.

A. Multilabel Classification in Remote Sensing

Early multilabel remote sensing image classification methods are mainly based on manually designed features, such as histogram of oriented gradient [47] and scale-invariant feature transforms [48]. These features are then fed into traditional machine learning methods, such as random forest [49] and support vector machines (SVMs) [50], for classification. Li et al. [13] proposed a multilabel SVM algorithm, where a variety of artificially designed features are first extracted and then input to an SVM classifier for classification. Koda et al. [27] utilized structured SVM to establish the relationship between labels and made use of the spatial context information of remote sensing images during classification. The above studies promoted the MLRSSC performance with machine learning approaches. However, the manually designed visual features make it difficult to express high-level semantic information, and the generalization ability of the models is unsatisfactory.

Recently, the prosperity of DL has triggered a new trend toward designing DL-based MLRSSC approaches. Specifically, the convolutional neural network (CNN) has been extensively applied in MLRSSC task as semantic feature learners. Yao et al. [51] designed deep representations of multimodal remote sensing data for land-cover classification. Li et al. [52] trained an interpretable deep unfolding architecture for anomaly detection of remote sensing images. Zeggada et al. [35] exploited the CNN to yield a powerful representation of remote sensing images and performed classification by combining a radial basis function-based deep neural network and a multilabeling block comprised of customized thresholding operations. Attributed to the powerful feature learning ability of DL techniques, the above methods significantly elevated the classification performance of MLRSSC task. However, the underlying class dependencies inherent in the studied task have not been taken into account.

B. Label Dependency Modeling

1) RNN-Based Methods: To model the interclass relationships, RNN [53] and LSTM [54] were introduced to the MLRSSC task. Wang et al. [55] put forward a CNN–RNN framework, where the RNN model is used to learn collaboratively image-label representations from CNN features, and recurrent LSTM neurons are employed to predict labels sequentially. Hua et al. [28] proposed to use bidirectional LSTM to implicitly learn label correlations. The CM–GM framework [14] projects visual features and label embeddings into a shared correlation space, which is fed to an LSTM subnetwork to facilitate the multilabel classification with label correlations. RNN-based models excel in modeling short-range sequence dependencies, however, have limited ability to represent graph-structured relations. A second disadvantage of the RNN is that the chain propagation manner easily leads to the accumulation of errors.

2) GNN-Based Methods: GNN is a graph-structured data-oriented network that aims to learn the spatial and topological relationships of graphs [56], [57]. Due to the solid capacity of the GNN in modeling relationships, a corpus of researchers proposed to use the GNN to model the label dependencies in the MLRSSC task.

Huang et al. [32] developed an MLRSSC-CNN-GNN framework, where a multilayer graph attention network was designed to explore the spatial relationships of remote sensing image scenes. Kang et al. [58] proposed to use graphs to model the label correlations among multiple remote sensing scenes. [32] designed a channel-spatial attention module to fuse multiscale features extracted with the CNN model. Then, class dependency information is drawn from the label co-occurrence matrix and incorporated into multiscale attention features. In [59], a multigranularity semantic grouping mechanism is leveraged to learn visual features, and a graph convolutional network (GCN) module is further used to directly extract the label correlations. Lin et al. [60] proposed to infer label relevance from datasets and knowledge graphs and combined semantic attention and label attention into the GCN. Zhu et al. [61] innovatively introduced dual-level semantic concept, wherein both scene-level and object-level labels are employed to guide the multilabel classification. Ma et al. [44] constructed a label correlation matrix by statistically analyzing the co-occurrence frequencies, where a label recognition GCN is followed by for scene classification.

C. Attention-Based Methods

Attention mechanism [62], which pushes the network to highlight the important information, attracted notable interests in the machine learning community. Specifically, self-attention-based Transformer [63] has shown powerful capability in long-range dependency modeling. In view of its strong representation ability, attention mechanism has been extended to the multilabel classification task [41], [64], [65], [66]. Hua et al. [29] proposed an attentional region extraction network to localize and re-coordinate attentional regions for consequent label inference. Tan et al. [39] presented a Transformer-driven semantic relation inference method which uses class activation mapping to locate regions of interests and a Transformer encoder to learn label relations. A masked attention Transformer was adopted in [40] to dynamically generate different masks based on the relevance of semantic

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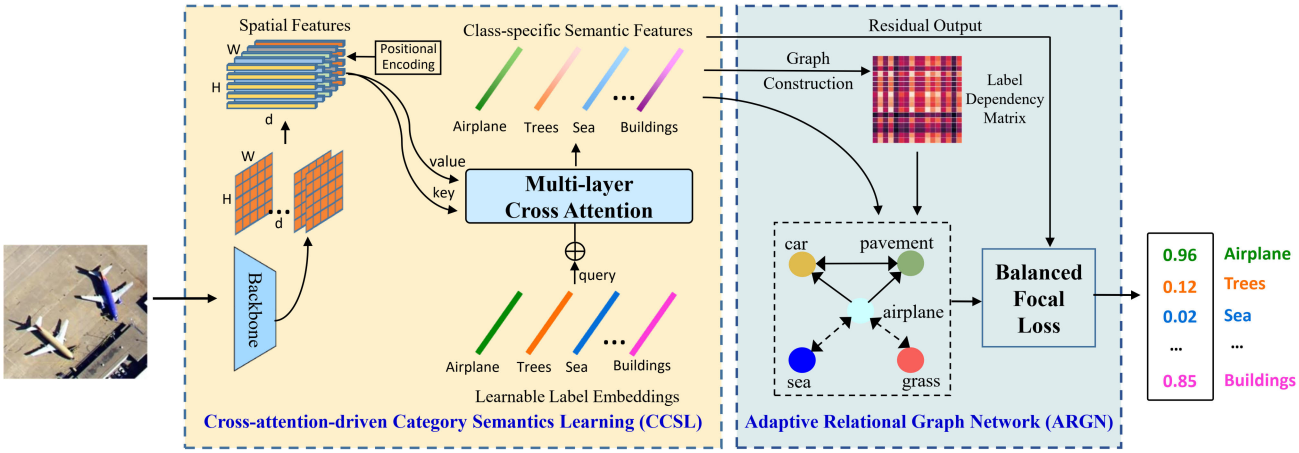


Fig. 3. Illustration of the cross-attention-driven adaptive graph relational network for MLRSSC task. Given an input image, the method first learns class semantics via a cross-attention mechanism with learnable label embeddings as queries and deep features extracted from CNN-based backbone networks as keys and values. With the learned class semantics, we next construct image-specific label dependency graphs, based on which node features are aggregated via the GCN, producing the final predictive result with the newly designed BFL function.

labels, allowing the attention mechanism to devote more focus on regions related to the target category.

Analyzing the related works in the MLRSSC task, we can draw the below conclusion.

1) The graph-based label dependency modeling approaches are all based on static graphs, neglecting the less-frequent co-occurring labels and class distribution discrepancy between training and test sets.
2) CNN-based and class attention-based approaches are commonly used as feature learners for MLRSSC tasks. However, neither of them involves the guidance of labels in the representation learning.
3) We do not discover any works which take into account the imbalance issues in the MLRSSC task.

The differences between our proposed method and existing works reside in three aspects.

1) Unlike static graphs, we employ dynamic graphs to model the interclass correlations, which adaptively establishes unique class relational graphs for each image.
2) The graph node representation is obtained via label embedding-driven cross-attentions. Specifically, the label embeddings are learnable parameters that are updated during the optimization, to accurately locate the class-wise semantic regions, elevating the descriptiveness of the learned features.
3) To our knowledge, the sample imbalance issue has never been dealt with in MLRSSC tasks. Our work provides an effective approach to bridge the gap.

III. Methodology

A. Overview

In this section, we will first formulate the studied problem and proceed by introducing the framework of our method.

The objective of the MLRSSC task is to predict the presence of multiple scene categories given an input remote sensing image. Assume there are C scene categories in total. Let x be an input image, the corresponding ground-truth label is denoted as $y \equiv \{y_1, y_2, \ldots, y_C\}$, $y_i \in \{0, 1\}$. We aim to construct a classifier f to perform the multilabel prediction by virtue of $\hat{y} = f(x)$.

The proposed CAGRN, as illustrated in Fig. 3, is composed of two modules, that is, the Cross-attention-driven Class Semantics Learning (CCSL) module, and Adaptive Relational Graph Network (ARGN) module. The CCSL module first uses a CNN-based backbone network to extract deep fine-grained feature maps and then learns discriminative class-specific semantic features by virtue of the cross-attention mechanism, with learnable label embeddings as queries, and the extracted spatial features as keys and values. With the class-wise semantic features as input, the ARGN module dynamically constructs image-specific interclass relational graphs and then performs feature aggregation through graph convolution to obtain the final predictions. Specifically, we design a new multilabel classification loss function, that is, BFL, to cope with the class imbalance and positive–negative imbalance challenges. The CCSL and ARGN modules combined with the BFL function will be detailed in Sections III-B–III-D, respectively.

B. Cross-Attention-Driven Class Semantics Learning

The discriminative ability of class semantics fundamentally determines the accuracy of scene identification and class correlation modeling. There are two key points to improve the expressive power of class semantics: 1) extraction of representative visual features and 2) accurate pooling of class-related semantic representations from the visual features. To achieve these two goals, we resort to CNN-based deep feature extraction and CCSL, which will be described as follows.

1) Deep Feature Extraction: Discriminative features are the prerequisite for the feature-label relation establishment and consequent multilabel classification.

In this work, we adopt ResNet50 [38] as the backbone network. Given an input image $M \in R^{3\times W_0\times H_0}$, the spatial feature embeddings $H \in R^{d_0\times W\times H}$ are generated with the backbone, where W_0, H_0, H, and W represent the widths and heights of

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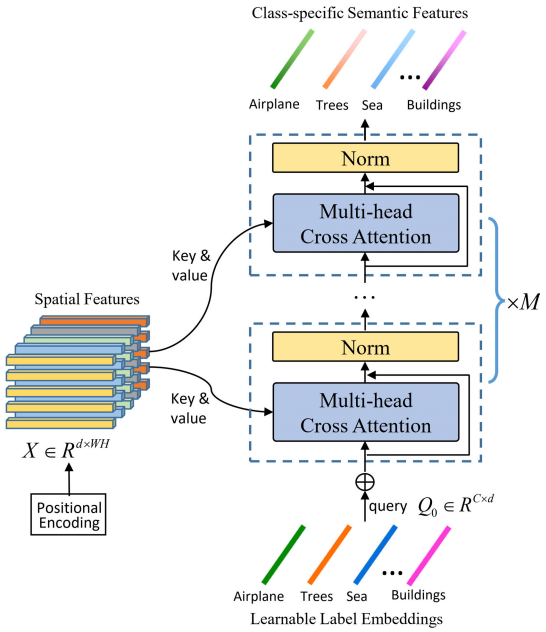


Fig. 4. Schematic illustration of the CCSL module.

the input image and the spatial feature map, respectively, and d_0 denotes the channel dimension of the spatial feature map. After that, we perform a channel transformation on the spatial feature map H to project the features from dimension d_0 to d, which is implemented by a 1×1 convolution. Then the pixel-level feature vector $X \in R^{d\times WH}$ in the feature map is obtained following:

$$X = f(g(H)) \tag{1}$$

where $f(\cdot)$ is the flatten operation and $g(\cdot)$ is the channel transformation operation.

2) Class Semantics Learning: The key to extracting discriminative class semantics is to precisely model the complex interactions between features and labels. The Transformer [42], [62] has proved to be a potent mechanism for capturing the intricate dependency information among variables. Inspired by this, we propose to exploit attention mechanism to model the complex interactions between features and labels. To be specific, the objective is achieved by encouraging the network to focus on regions of interest corresponding to each category in the spatial feature map, thereby producing content-aware class representations.

Fig. 4 illustrates the CCSL process. As shown in the figure, the label embeddings $Q_0 \in R^{C\times d}$ are fed to the cross-attention layer as queries, and the pixel-level spatial features $X \in R^{d\times WH}$ combined with the positional encodings act as keys and values. By computing the similarities between the label embeddings and features at each spatial location, we can obtain the class-wise attention maps. The spatial feature maps are further multiplied by the yielded attention maps, highlighting features specific to each category. It should be noted that the label embeddings are learnable parameters which are updated during the learning process. To obtain more discriminative features, we use M standard cross-attention layers to update the queries. The ith ($i > 1$) cross-attention layer updates query q_i from the output of its previous layer following:

$$\begin{aligned} Q_{i-1} &= q_{i-1}W_{i-1}^{Q}, \{K_{i-1}, V_{i-1}\} = XW_{i-1}^{\{K,V\}} \\ \tilde{q}_i &= \text{CrossAttn}(Q_{i-1}, K_{i-1}, V_{i-1}) \\ &= \text{Softmax}\left(\frac{Q_{i-1}{K_{i-1}}^{T}}{\sqrt{d}}\right)V_{i-1} \\ q_i &= \text{Norm}(\tilde{q}_i) \end{aligned} \tag{2}$$

where $W_{i-1}^{Q}, W_{i-1}^{K}, W_{i-1}^{V}$ are three learned weight matrices in the $(i-1)$th layer, $Q_{i-1}, K_{i-1}, V_{i-1}, \tilde{q}_i$ are all intermediate variables, CrossAttn$(\cdot)$ and Norm$(\cdot)$ denote the cross-attention and normalization operations, respectively, and $\sqrt{d}$ is a scaling factor to avoid the variance effect caused by the dot product.

In this paragraph, we provide an intuitive description of how the cross-attention mechanism establishes the relationship between spatial feature maps and label embeddings. Let $q^c \in R^d$ denote the learnable label embedding vector of category c, we calculate the attention score between q^c and each possible pixel-level feature vector. Then, the normalized attention score is leveraged to weigh each pixel-level feature vector to obtain the final feature vector $v^c \in R^d$ for category c. The label embeddings of all categories are obtained by repeating the above process over all categories, which will be further exploited as the input of the next cross-attention layer. Finally, category nodes $V = \{v^1, v^2, \ldots, v^C\} \in R^{C\times d}$ are obtained, acting as input for the subsequent dynamic graph construction. In summary, the essence of the cross-attention is to pool features of interest from the spatial feature map using categorical embeddings. Then through iterative parameter optimization, representative and discriminative class semantics can be learned with contextualized information from the input image progressively incorporated.

C. Adaptive Relational Graph Network

Given the representative class semantics learned in the CCSL module, the ARGN module aims to construct relational graphs, to explicitly model the scene label dependencies, and further aggregate node representations. In light of the natural advantages of graphs in modeling relations, we leverage the GCN to capture the underlying correlations among scene category labels. It should be emphasized that our proposed graphs are different from conventional static graphs which are built in prior based upon category co-occurrence statistics or similarities among label word embeddings, and are shared by all images in the dataset. Instead, we dynamically build image-specific graphs, which means that the graphs are adaptively constructed based on the image-specific class semantics. There are two essential procedures for the graph-based feature optimization, that is, adaptive relational graph construction and graph convolutional propagation, which will be delineated in Sections III-C1 and III-C2, respectively.

1) Adaptive Relational Graph Construction: With the learned class semantic features as input, the objective of the graph construction is to produce a relational graph, that is, label dependency matrix, which is the prerequisite process for the consequent feature aggregation.

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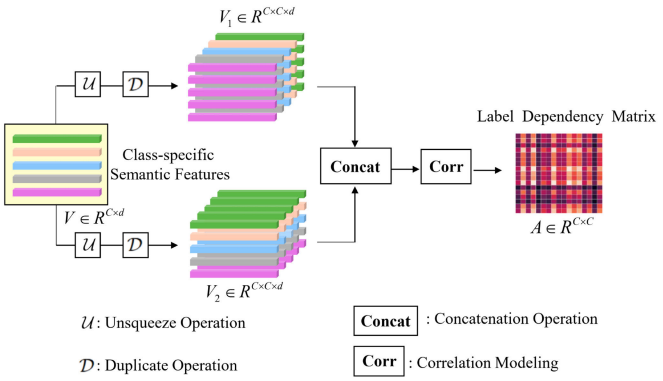


Fig. 5. Schematic illustration of the proposed adaptive category relational graph construction.

Fig. 5 illustrates our proposed adaptive relational graph construction process. We first extend the size of the 2-D semantic feature set V to sizes $C \times 1 \times d$ and $1 \times C \times d$, respectively, via unsqueeze operation, and duplicate the features in the newly added dimension, yielding V_1 and V_2 with the same size of $C \times C \times d$. V_1 and V_2 are next concatenated to a feature vector V_c with size $C \times C \times 2d$. Consequently, we compute the correlation score among each category pair following:

$$S = \text{Squeeze}(\text{Conv}(\tanh(V_c)))$$
$$A = \text{Sigmoid}(S) \tag{3}$$

where activation function $\tanh(\cdot)$, convolution operation $\text{Conv}(\cdot)$ with kernel size $2d \times 1$, and compression on the third dimension realized via $\text{Squeeze}(\cdot)$ operation are sequentially implemented. Finally, the $\text{Sigmoid}(\cdot)$ function is performed on the obtained correlations scores S to generate the label dependency matrix $A \in R^{C\times C}$.

It is important to underline that the label dependency matrix A is uniquely determined by the input class-specific semantic feature V and differs among diverse samples. Therefore, during inference, the label dependency matrix A is adaptively and dynamically generated for each test image, which intrinsically alleviates the over-fitting risk and correlation bias that static graph brings about.

2) Graph Convolutional Propagation: With the class semantic node features V as input, the GCN operates by propagating information between nodes based on the adaptive label dependency matrix A and a learnable graph weight matrix $W_A \in R^{d\times d}$. For the label dependency matrix A reveals the relations among the graph nodes, the node representation update therefore involves the aggregation of the influences from relational nodes. To enhance the stability of the model, we further integrate a residual connection branch weighted by matrix $W_V \in R^{d\times d}$ to the architecture. Under this design, the graph node aggregation process can then be expressed as

$$Z = \sigma(AVW_A) + VW_V \tag{4}$$

where $\sigma(\cdot)$ denotes the nonlinear activation function which takes the form of $\text{ReLU}(\cdot)$ activation function and Z is the updated class semantic representation. Both W_A and W_V are learnable weight matrices.

D. Classification and BFL

The combination of class semantic features Z and a residual connection branch from the CCSL module is used as input for the final classification. The binary classifier consists of a fully connected layer and an activation layer. Thereafter, the final output probability $p = \{p_1, p_2, \ldots, p_C\}$ can be obtained following:

$$p = \text{Softmax}(\text{FC}(\beta Z + \lambda V)) \tag{5}$$

where $\text{FC}(\cdot)$ represents the fully connected layer, and the activation layer is taken as the $\text{Softmax}(\cdot)$ function. β and λ are two weight coefficients, which are set to 0.3 and 0.7 in this work, respectively.

To alleviate the problem of sample imbalances, we propose the BFL function, which is an extension of the traditional FL function designed originally for object detection tasks [46]. The total loss function can be written as

$$\mathcal{L} = -\sum_{i=1}^{N}\sum_{c=1}^{C}\left(1-p_i^c\right)^{\gamma^c} y_i^c \log p_i^c + \left(p_i^c\right)^{\gamma^c}\left(1-y_i^c\right)\log\left(1-p_i^c\right) \tag{6}$$

where N and C denote the training sample number and the class number, respectively, and y_i^c and p_i^c are the ground-truth label and output probability for class c of the ith sample, respectively.

In (6), γ^c is a newly defined balancing factor, which is composed of a class-agnostic parameter γ_b and a class-specific component. The definitions are as below:

$$\gamma^c = \gamma_b - s \times g^c \tag{7}$$

where γ_b indicates the class-agnostic factor under the sample-balanced scenario which guarantees the basic behavior of the multilabel classification. g^c denotes the ratio of positive samples to negative samples of class c in the training set. s is a scaling parameter that adjusts the influence of class imbalance in BFL.

It is worth mentioning that different from the original FL, BFL tackles the positive–negative imbalance of each class independently. Different g_c yields different γ^c parameter. It can be inferred from (7) that with the decrease of g_c value, γ^c will be increased, which means that more focus will be placed on hard positive samples. Via the above designs, the positive–negative imbalance and class imbalance can be handled simultaneously. FL is a special case of BFL, which can be obtained by setting γ_b to the same γ value as FL and s to 0.

IV. Experimental Results and Discussion

We validate the proposed algorithm on three benchmark datasets and analyze its performance in this section. We first introduce the experimental data and settings in Section IV-A. Then, we compare the performance of CAGRN with nine competitors in Section IV-B. Taking the UCM multilabel dataset, for example, we conduct ablation studies to verify the effectiveness of the key components of CAGRN in Section IV-C. In Section IV-E, we proceed to study the impact of parameters. Finally, we visualize the activation maps and the yielded relational graphs in Section IV-F.

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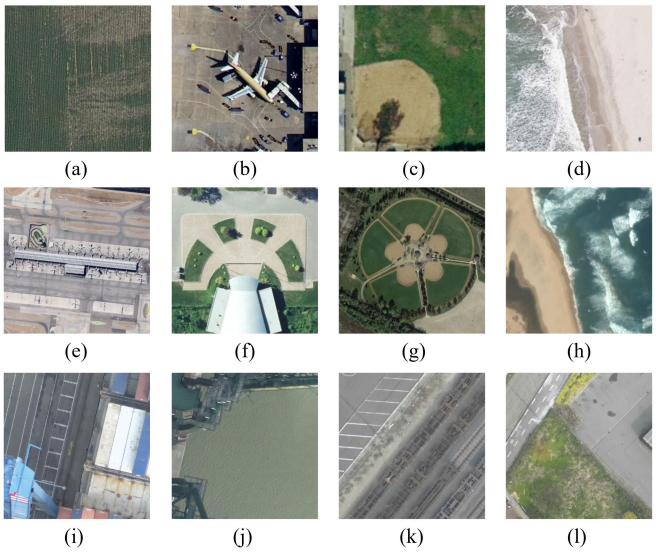


Fig. 6. Examples of UCM, AID, and DFC15 multilabel datasets and their corresponding ground-truth labels. (a)–(d) UCM dataset. (e)–(h) AID dataset. (i)–(l) DFC15 dataset. (a) *Field*. (b) *Airplane*, *buildings*, *cars*, and *pavement*. (c) *Bare soil*, *grass*, *pavement*, and *trees*. (d) *Sand* and *sea*. (e) *Airplane*, *bare soil*, *buildings*, *cars*, *grass*, and *pavement*. (f) *Buildings*, *grass*, *pavement*, and *trees*. (g) *Bare soil*, *buildings*, *grass*, *pavement*, and *trees*. (h) *Sand* and *sea*. (i) *Impervious* and *clutter*. (j) *Impervious*, *water*, and *clutter*. (k) *Impervious* and *clutter*. (l) *Impervious*, *vegetation*, *buildings*, and *trees*.

A. Experimental Data and Settings

1) Experimental Data: We will give the details of three benchmark experimental datasets, that is, UCM, AID, DFC15 multilabel datasets, in the following paragraphs, respectively.

a) UCM multilabel dataset: The UCM multilabel dataset was released by the U.S. Geological Survey's National Map [45]. The original dataset comprises 2100 images categorized into 21 classes. There are 17 scene labels in total, including *airplanes*, *bare soil*, *buildings*, *cars*, *chaparral*, *court*, *dock*, *field*, *grass*, *mobile home*, *pavement*, *sand*, *sea*, *ship*, *tanks*, *trees*, and *water*. Each image in the dataset was relabeled with one or more of the above scene labels in [59], where the ground-truth data is available at http://bigearth.eu/datasets. The image size is 256 × 256 × 3 and the spatial resolution is 0.3 m. Fig. 6(a)–(d) shows several example images and their corresponding ground-truth labels in the dataset.

b) AID multilabel dataset: The original AID dataset [67] was provided by Wuhan University, which includes 3000 aerial images with 17 scene labels which are consistent with the UCM dataset. In [29], the dataset was reproduced for multilabel scene classification, where the multilabel annotations are available at https://github.com/Hua-YS/AID-Multilabel-Dataset. The image size is 600 × 600 × 3. Fig. 6(e)–(h) shows four example images from the AID dataset.

c) DFC15 multilabel dataset: The DFC15 dataset was first published in 2015 IEEE GRSS Data Fusion Contest [68], and further annotated with multiple labels in [28]. The dataset is available at https://github.com/Hua-YS/DFC15-Multilabel-Dataset. It contains 3342 images with size 256 × 256 × 3. The dataset includes eight classes, which are *building*, *boat*, *car*, *clutter*, *impervious*, *water*, *vegetation*, and *trees*, respectively. Four example images are displayed in Fig. 6(i)–(l).

2) Experimental Setting: Following [39], [44], the images in the UCM and DFC15 datasets keep the original size 256 × 256, and the images in the AID dataset are resized to 512 × 512. Three data augmentation strategies are utilized in sequence, which are random horizontal flips, random vertical flip, and color jitter, respectively. To ensure the fairness of comparison with other competitors, we adopt the same data division as [59], where 80% of images in each scene class are randomly selected as a training set, and the remaining 20% of images are used as a test set. We train the model using the Adamw optimizer with the weight decay setting to 0.01. The learning rate is set to 0.0001 initially and decays by 0.1 every 5 epochs starting from the 10-*th* epoch. We train our model for 30 epochs in total with a batch size of 32. There are a total of eight cross-attention heads, and the feature dimension is set to 2048. We use two cross-attention layers for UCM and DFC15 datasets, and four cross-attention layers for the AID dataset. In experiments, γ_b is set to 2, while s is set to 0.1 for AID and UCM datasets, and 0.05 for the DFC15 dataset. All the experiments are implemented in PyTorch framework on an OmniSky SCS4450-G2 workstation with one GeForce RTX A5000 GPU and 24 GB memory.

3) Evaluation Metrics: Following [44], we adopt two groups of evaluation metrics, which are overall $F1$ score OF1, precision OP, recall OR, and per-class $F1$ score CF1, precision CP, recall CR, respectively, which are defined as follows:

$$\mathrm{OF1} = \frac{2 \times \mathrm{OP} \times \mathrm{OR}}{\mathrm{OP} + \mathrm{OR}}, \tag{8}$$

$$\mathrm{OP} = \frac{1}{N_{ts}} \sum_{i=1}^{N_{ts}} \frac{\mathrm{TP}_e}{\mathrm{TP}_e + \mathrm{FP}_e} \tag{9}$$

$$\mathrm{OR} = \frac{1}{N_{ts}} \sum_{i=1}^{N_{ts}} \frac{\mathrm{TP}_e}{\mathrm{TP}_e + \mathrm{FN}_e} \tag{10}$$

$$\mathrm{CF1} = \frac{2 \times \mathrm{CP} \times \mathrm{CR}}{\mathrm{CP} + \mathrm{CR}} \tag{11}$$

$$\mathrm{CP} = \frac{1}{C_{ts}} \sum_{i=1}^{C_{ts}} \frac{\mathrm{TP}_l}{\mathrm{TP}_l + \mathrm{FP}_l} \tag{12}$$

$$\mathrm{CR} = \frac{1}{C_{ts}} \sum_{i=1}^{C_{ts}} \frac{\mathrm{TP}_l}{\mathrm{TP}_l + \mathrm{FN}_l} \tag{13}$$

where TP_e (example-based true positive) and FP_e (example-based false positive), respectively, represent the numbers of correctly predicted positive labels and unrecognized positive labels, while FN_e (example-based false negative) denotes the number of incorrectly predicted negative labels in an image. TP_l (label-based true positive) and FP_l (label-based false positive) denote the numbers of correctly predicted positive labels and unrecognized positive labels in a class, while FN_l (label-based false negative) indicates the number of incorrectly predicted negative labels in the class. Besides, N_{ts} and C_{ts} denote the number of images and classes in the test set, respectively. In our experiments, a label is judged as positive when its corresponding confidence score is larger than 0.5.

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TABLE I
PERFORMANCE COMPARISON WITH STATE-OF-THE-ART METHODS ON THE UCM MULTILABEL DATASET (%). THE BEST RESULTS ARE MARKED AS BOLD

Method	OP	OR	OF1	CP	CR	CF1
CA-BiLSTM [28]	77.94	89.02	81.47	86.12	84.26	85.18
AL-RN [29]	88.81	87.07	86.76	92.33	85.95	89.03
CM-GM [14]	81.60	89.65	81.58	88.57	85.20	86.19
MSGM [33]	83.86	85.48	84.54	89.98	85.07	87.46
ML-CG [60]	81.34	89.94	85.42	88.53	89.27	88.90
SR-Net [39]	87.96	89.40	88.67	93.52	91.51	92.50
GRN [58]	91.98	92.83	91.31	-	-	-
Huang [32]	90.54	92.98	91.74	93.73	92.75	93.23
LD-GCN [44]	93.21	88.39	90.73	96.31	93.24	94.70
Our CAGRN	**93.75**	**93.01**	**93.38**	**96.74**	**95.34**	**96.04**

B. Comparisons With Other Methods

To evaluate the performance of our proposed method, we compare it with nine state-of-the-art multilabel classification approaches, which are briefly described below.

1) *CA-BiLSTM [28]:* This method uses a class activation mapping layer to capture class-specific representations and then employs a bidirectional LSTM module to model the label correlations for multilabel classification.
2) *AL-RN [29]:* In this method, an attentional region extraction module is designed for feature extraction, and a label relational inference module is utilized for class dependency modeling.
3) *GRN [58]:* A scalable neighbor discrimination and BCE loss function is devised in this work, which aims to model the relationship between scenes through network embedding graph structures.
4) *Huang [32]:* The method exploits a channel-spatial attention mechanism to fuse multiscale features extracted with the CNN. A statistically computed label co-occurrence matrix is employed to model the label dependencies.
5) *ML-CG [60]:* This method uses the GNN and knowledge graph to infer the correlation between labels, based on semantic features extracted with an image decoupling module which is composed of a ResNet block, a low-rank bilinear pooling layer, and an attention block.
6) *SR-Net [39]:* This work designs a semantic-sensitive module to generate category representations and a Transformer encoder to model the interclass dependencies.
7) *MSGM [33]:* The authors resort to the multigrained semantic grouping mechanism to learn visual features and a GCN module to capture the label correlations.
8) *CM-GM [14]:* This method establishes the mapping between visual embeddings and labels in a shared correlation space, effectively using label dependencies to improve the capabilities of LSTM predictors.
9) *LD-GCN [44]:* This is a label-driven GCN which uses the co-occurrence frequencies of different categories to construct the label correlation matrix.

1) UCM Multilabel Dataset: Table I illustrates the numerical results of ten compared methods. From the table, we can conclude that our proposed CAGRN achieves the best classification performance over all six evaluation metrics, which outperforms the second-best method, that is, LD-GCN, by 2.65% and 1.34% on OF1 and CF2, respectively. We notice that CA-BiLSTM and CM-GM yield 81.47% and 81.58% on OF1 metric and 85.18% and 86.19% on CF1 metric, which are the lowest among all compared methods. This may be because both methods leverage LSTM to model the label dependencies, whose chain propagation manner exhibits inferior ability in modeling the complicated graph-structured relationships. By contrast, our proposed method capitalizes on graph to model the label dependencies. Specifically, the constructed graphs are image-specific, which are capable of adaptively representing the interclass correlations of each image, further facilitating the improvement of the multilabel classification performance.

2) AID Multilabel Dataset: The classification results of all compared methods on the AID multilabel dataset are displayed in Table II. It can be discovered from the table that CAGRN obtains 93.55% and 82.48% on OF1 and CF1 Metrics, respectively, which are both the highest among all compared methods. Specifically, it surpasses the second highest OF1 and CF1 values by 2.52% and 4.69%, respectively. It is noted that AL-RL and SR-Net perform inferiorly on this dataset. This can be attributed to the fact that both approaches employ global attention mechanism to extract categorical feature embeddings, without explicitly modeling the relationships between features and labels. Different from them, our proposed CAGRN uses learnable label embeddings as queries to drive the semantic feature learning, which explicitly establishes feature-label connections, producing representative graph node features for the final graph-based category inference.

3) DFC15 Multilabel Dataset: Table III reports the classification results of all compared methods over the DFC15 multilabel dataset. It can be observed from the table that our proposed CAGRN method again yields the best classification performance over OF1 and CF1 evaluation metrics. Specifically, CAGRN outperforms CA-BiLSTM by 13.61% and 20.74% on OF1 and CF1 metrics, respectively, which precisely demonstrates the advantage of dynamic graphs in modeling label correlations over the LSTM method.

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TABLE II
PERFORMANCE COMPARISON WITH STATE-OF-THE-ART METHODS ON THE AID MULTILABEL DATASET (%). THE BEST RESULTS ARE MARKED AS BOLD

Method	OP	OR	OF1	CP	CR	CF1
AL-RN [29]	91.00	88.95	88.72	80.81	71.12	75.66
SR-Net [39]	89.42	90.52	89.97	**87.24**	**82.25**	-
GRN [58]	92.79	91.08	90.95	-	-	-
Huang [32]	91.88	91.44	91.03	81.37	74.60	77.79
LD-GCN [44]	92.81	89.06	90.93	85.14	71.22	77.49
Our CAGRN	**95.07**	**92.07**	**93.55**	87.05	78.37	**82.48**

TABLE III
PERFORMANCE COMPARISON WITH STATE-OF-THE-ART METHODS ON THE DFC15 MULTILABEL DATASET (%). THE BEST RESULTS ARE MARKED AS BOLD

Method	OP	OR	OF1	CP	CR	CF1
CA-BiLSTM [28]	91.93	79.12	83.65	94.35	62.35	75.08
MSGM [33]	94.61	92.71	93.65	91.42	90.70	91.06
GRN [58]	96.53	95.95	95.80	-	-	-
ML-CG [60]	95.68	96.09	95.89	94.28	93.63	93.96
LD-GCN [44]	96.32	**97.09**	96.74	95.43	**95.50**	95.54
Our CAGRN	**98.56**	96.00	**97.26**	**97.98**	93.75	**95.82**

TABLE IV
COMPONENT EFFECTIVENESS EVALUATION OF THE PROPOSED METHOD ON THE UCM DATASET

Models	OP	OR	OF1	CP	CR	CF1
ResNet-50	93.10	90.44	91.75	96.56	93.52	95.02
ResNet-50+CCSL	93.62	92.08	92.84	96.59	94.58	95.57
CAGRN	**93.75**	**93.01**	**93.38**	**96.74**	**95.34**	**96.04**

C. Ablation Study

In this set of experiments, taking UCM multilabel dataset as an example, we carry out ablation studies to demonstrate the effectiveness of each component of our proposed CAGRN method, which are the cross-attention-driven category semantics learning (CCSL) module, the ARGN module and the BFL function, respectively.

To verify the effectiveness of CCSL and ARGN modules, we performed three groups of experiments with each component successively added to the baseline ResNet-50 method. Table IV tabulates the classification results over six evaluation metrics. It can be obviously discovered from the table that with CCSL and ARGN progressively added to the baseline, all values of the six metrics increase compared with their counterparts without the modules, which exactly verify the effectiveness of each component.

To demonstrate the effectiveness of BFL function in dealing with the sample imbalance issues, we conducted another three groups of experiments with BCE, FL, and BFL as the classification loss functions. Table V exhibits the numerical classification accuracy on all classes and the OF1 and CF1 metrics. It can be revealed from the table that BFL obtains the highest values on OF1 and CF1 metrics, and the best classification accuracy of 12 classes out of 17. Specifically, for most tail classes, such as *court*, *sand*, and *trees*, BFL achieves the highest classification accuracy (denoted as bold in the table), which demonstrates that our designed loss function is effective in dealing with sample imbalance issues.

D. Analysis on Space and Time Complexities

Apart from classification performance, computational space and time complexities are also important indicators of the applicability of the methods. In this paper, we choose a number of parameters and floating-point operations (FLOPs) as the space and time complexity evaluation metrics, respectively. Table VI tabulates the values of the selected metrics with CCSL and ARGN modules progressively added to the network. As shown in the table, the CCSL module brings about 24.06 MB parameters and 1.64G FLOPs increase, while the ARGN module introduces 8.01 MB parameters and 0.07G FLOPs. To conclude, the parameters and FLOPs of the proposed method are augmented by 35.7% and 31.6% compared with the baseline network, which are moderate increases concerning the state-of-the-art classification performance it achieves.

E. Parameter Analysis

In this section, we analyze the impacts of different parameters on the classification performance. We, respectively, study

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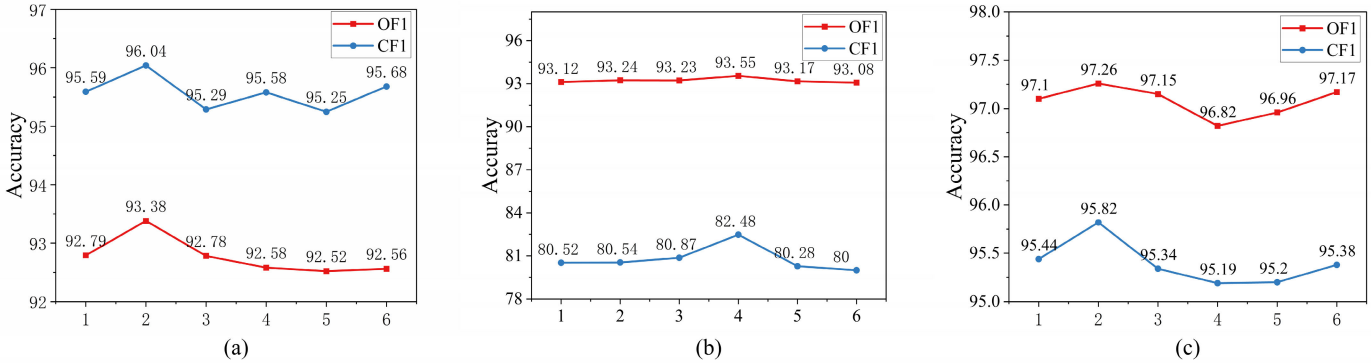


Fig. 7. Classification performance with different numbers of cross-attention layers on three datasets. (a) UCM dataset. (b) AID dataset. (c) DFC15 dataset.

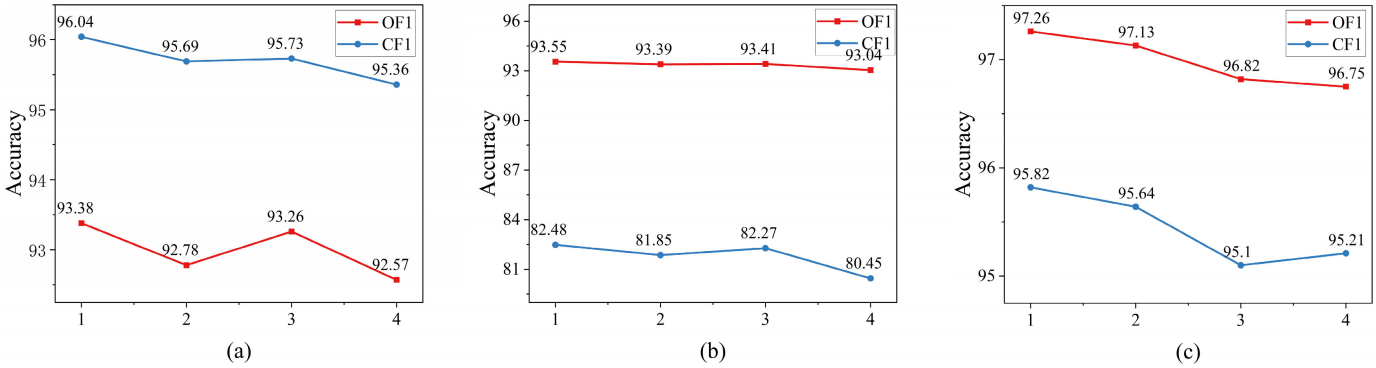


Fig. 8. Classification performance with different numbers of graph convolutional layers on three datasets. (a) UCM dataset. (b) AID dataset. (c) DFC15 dataset.

TABLE V
CLASSIFICATION RESULTS WITH DIFFERENT LOSS FUNCTIONS ON THE UCM DATASET

Classes	BCE	FL	BFL
Airplane	**100.0**	**100.0**	**100.0**
Bare soil	81.72	83.87	**84.81**
Buildings	92.96	94.04	**94.12**
Cars	**90.96**	89.53	89.77
Chaparral	**97.67**	**97.67**	**97.67**
Court	95.00	95.24	**97.56**
Dock	**100.0**	**100.0**	**100.0**
Field	95.00	**95.24**	95.00
Grass	90.86	**92.35**	92.06
Mobile home	**97.56**	**97.56**	**97.56**
Pavement	**96.00**	95.42	95.27
Sand	97.44	94.74	**98.28**
Sea	**100.0**	**100.0**	**100.0**
Ship	**100.0**	**100.0**	**100.0**
Tanks	94.74	**100.0**	**100.0**
Trees	90.82	92.89	**93.00**
Water	**98.77**	**98.77**	97.56
OF1	92.54	93.07	**93.38**
CF1	95.30	95.73	**96.04**

TABLE VI
SPACE AND TIME COMPLEXITIES OF THE PROPOSED METHOD

Models	Parameters (MB)	FLOPs (G)
ResNet-50	89.81	5.41
ResNet-50+CCSL	113.87	7.05
CAGRN	121.88	7.12

the influences of the number of cross-attention layers and the number of graph convolutional layers in Sections IV-E1 and IV-E2, respectively.

1) Cross-Attention Layer: To investigate the influence of various numbers of cross-attention layers on the classification performance, we conduct experiments with {1, 2, 3, 4, 5, 6} as the cross-attention layer candidate set. Fig. 7 illustrates the OF1 and CF1 values with respect to different cross-attention layers on three datasets. Analyzing Fig. 7, we can find that increasing the number of cross-attention layers does not always bring gains in the classification performance. For UCM and DFC15 datasets, the best classification performance is achieved with cross-attention layer number setting to 2, while for the AID dataset, the best results are obtained with four cross-attention layers.

2) Graph Convolutional Layer: To study the impact of the number of graph convolutional layers on the classification performance, we tested on four-layer depth numbers, which are 1, 2, 3, and 4, respectively. We illustrate the numerical results with regard to the number of graph convolutional layers in Fig. 8. It can be inferred from the figure that with the increase of the layer number, the classification performance basically deteriorates, which can be explained by the presence of overfitting in large model depth scenarios. Comprehensively analyzing the experimental results, we use 1 as the graph convolutional layer setting as default.

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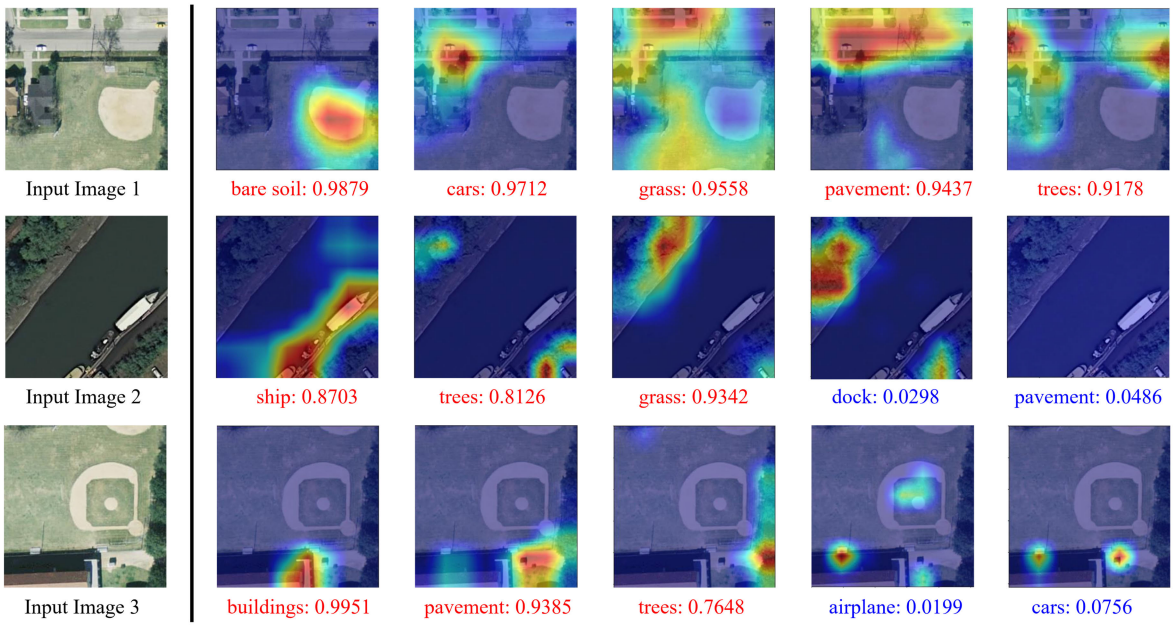


Fig. 9. Visualization of class-specific activation maps on three examples in the UCM dataset. The red text below the subfigure indicates the corresponding label is positive, while the blue text indicates a negative label.

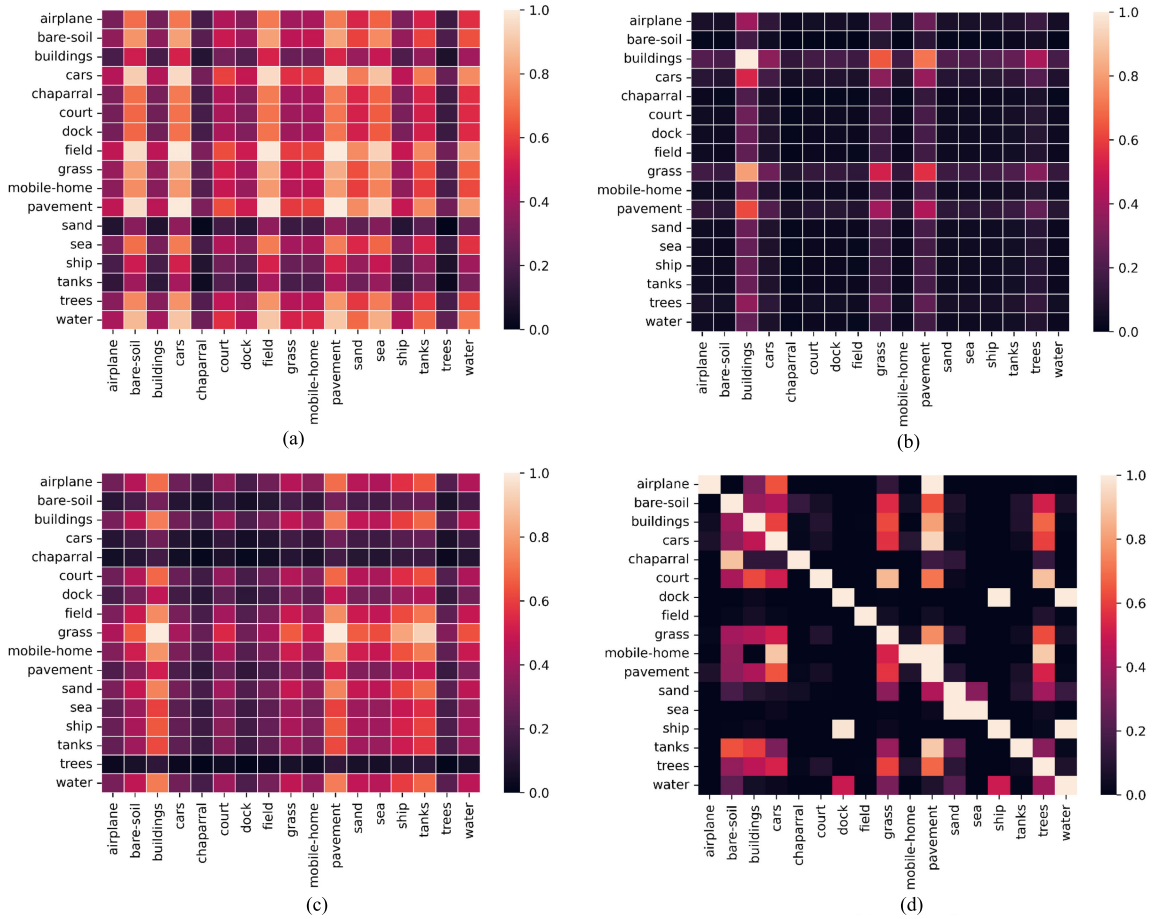


Fig. 10. Visualization of corresponding dynamic graphs of examples in Fig. 9 and the static graph in the UCM dataset. (a) Dynamic graph of input image 1. (c) Dynamic graph of input image 3. (b) Dynamic graph of input image 2. (d) Static graph.

F. Visualization

To intuitively explain why CCSL and ARGN modules work, we visualize the class-specific activation maps and dynamic graphs in this section.

1) Visualization of Class-Specific Activation Maps: Taking three images from the UCM dataset as examples, we visualize the activation maps generated with our proposed CAGRN in Fig. 9. Each row gives the input image and class-specific

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activation maps, with the corresponding labels and scores listed below. The labels and scores colored with red denote the positive labels, while the blue-colored labels and scores indicate the negative ones. Comparing the input images with the activation maps, we can discover that for classes that are present in the ground-truth labels, CAGRN can correctly locate the regions of interest. For example, the *bare soil* and *cars* in the first row, the *ship* and *trees* in the second row, and the *buildings* and *trees* in the third row are precisely highlighted. Whereas, for classes that are absent in the image, such as the *dock* and *pavement* in the second row and the *airplane* and *cars* in the third row, the corresponding activation maps exhibit low scores, which signifies low presence probabilities of these scene classes. The activation maps combined with the scores demonstrate that the CCSL module is capable of accurately locating target positions, which implies that discriminative class-wise semantic features are obtained in the process.

2) Visualization of Relational Graphs: As displayed in Fig. 10(a)–(c), we visualize the dynamic graphs corresponding to three example images in Fig. 9, to illustrate the label correlations the ARGN module has learned. For easy comparison, we exhibit the statistically derived static graph in Fig. 10(d) as well. The most essential point that needs to be emphasized is that our dynamic graphs are image-specific, which means that CAGRN generates a unique relational graph for each input image, while all images share one label dependency matrix in the static graph design. One shortcoming of the static graph is that it inevitably brings about frequency bias when it is applied to each specific image. Observing the static graph in Fig. 10, we can find a very high relational value between *ship* and *dock* categories. However, these two classes are not always present at the same time. Taking the input image 2, for example, it only contains *ship* target with *dock* target absent. Under this circumstance, if the static graph is exploited in the final category inference, the erroneous high *ship–dock* correlation would mislead the correct classification of the *dock* class. Nevertheless, the proposed dynamic graph is built based on class-specific semantic representations. Only classes appearing together in one single image would be assigned with a high dependency value. For *dock* class is absent in input image 2 (the score is 0.0298 as shown in Fig. 9), the *ship–dock* correlation score is extremely low, which will surely benefit the classification of the *dock* class. This phenomenon happens in the *cars–pavement* class pair for input image 3 as well.

V. Conclusion

In this article, we presented a cross-attention-driven adaptive graph relational network called CAGRN for an MLRSSC task. The advantages of the proposed method lie in three points.

1) CAGRN uses dynamic graphs to model label dependencies, adaptively constructing image-specific interclass relational graphs.

2) Cross-attention-driven feature learning explicitly extracts discriminative class-wise semantic representations, with label embeddings as direct queries.

3) The BFL effectively solves the class imbalance and positive–negative imbalance issues. Comprehensive experiments on three benchmark remote sensing multilabel datasets demonstrate the superiority of our proposed CAGRN method, which achieves state-of-the-art performance in the MLRSSC task.

In the future, we would like to probe into below directions: 1) designing few-shot learning-based MLRSSC methods, especially self-supervised contrastive and generative learning approaches, to tackle the longstanding label scarcity issue; and 2) devising a multiscale feature learning approach to address the object size divergence issue in remote sensing images.

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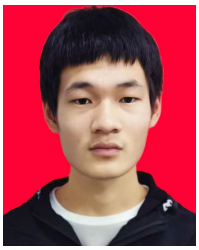
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Haixia Bi (Member, IEEE) received the B.S. and M.S. degrees in computer science and technology from the Ocean University of China, Qingdao, China, in 2003 and 2006, respectively, and the Ph.D. degree in computer science and technology from Xi'an Jiaotong University, Xi'an, China, in 2018.

She was a Post-Doctoral Research Fellow with the University of Derby, Derby, U.K., from 2018 to 2019, and the University of Bristol, Bristol, U.K., from 2019 to 2021. She is currently an Associate Professor with the School of Information and Communications Engineering, Xi'an Jiaotong University. Her research interests include machine learning and remote sensing image processing.

Dr. Bi was a recipient of the Best Reviewer Award of IEEE GEOSCIENCE AND REMOTE SENSING LETTERS in 2019 and 2021, respectively. She serves as a Guest Editor for *Remote Sensing Special Issue*.



Honghao Chang received the B.S. degree in information and communication engineering from Xi'an Jiaotong University, Xi'an, China, in 2023, where he is currently pursuing the Ph.D. degree with the School of Information and Communications Engineering.

His research interests include remote sensing image processing and multimodal learning.



Xiaotian Wang received the M.S. and Ph.D. degrees from Northwestern Polytechnical University, Xi'an, China, in 2016 and 2020, respectively.

He is an Associate Professor with Northwestern Polytechnical University. His research interests include computer vision and remote sensing image processing, especially object detection and tracking.



Danfeng Hong (Senior Member, IEEE) received the Dr.-Ing degree (summa cum laude) from the Signal Processing in Earth Observation (SiPEO), Technical University of Munich (TUM), Munich, Germany, in 2019.

Since 2022, he has been a Full Professor with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China. His research interests include artificial intelligence, multimodal vision, foundation models, and earth observation.

Dr. Hong received the Jose Bioucas Dias Award for recognizing an outstanding paper at WHISPERS in 2021, the Remote Sensing Young Investigator Award in 2022, the IEEE GRSS Early Career Award in 2022, the MIT Technology Review and DeepTech China's Intelligent Computing Innovators Award in 2024, and a Highly Cited Researcher (Clarivate Analytics/Thomson Reuters) since 2022. He is an Associate Editor for IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING (TGRS) and IEEE TRANSACTIONS ON IMAGE PROCESSING (TIP) and the Editorial Board Member of Information Fusion, *ISPRS Journal of Photogrammetry and Remote Sensing*.

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